

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

show ip route ospf

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? Serial0/0/1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? 65

Does R2 show up as an OSPF neighbor on R1? No

Does R2 show up as an OSPF neighbor on R3? No

What does this information tell you?

When an interface is set to passive, it stops transmitting OSPF Hello packets. As a result, the router cannot establish or maintain neighbor adjacencies through that interface, preventing any dynamic routing updates from being exchanged over that specific link.

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

R2(config)# router ospf 1

R2(config-router)# no passive-interface s0/0/1

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? Serial0/0/1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

65

Is R2 listed as an OSPF neighbor to R3? Yes

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

Please ensure reference bandwidth is consistent across all routers.

Why would you want to change the OSPF default reference-bandwidth?

____ Because the default reference bandwidth in OSPF is set to 100 Mbps, any link running at 100 Mbps or faster is assigned a default cost of 1. Modifying this reference value allows OSPF to accurately differentiate between high-speed links and calculate better optimal paths.

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

____ The cost is calculated using the formula: Reference Bandwidth / Interface Bandwidth. The total cost is the sum of all outgoing interface costs along the path.

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

____ The new cost for the serial link becomes 781. This happens because we manually changed the interface bandwidth to 128 kbps. OSPF recalculates the cost based on this new value, significantly increasing the accumulated cost.

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

____ By manually configuring the cost of interface S0/0/1 to 1565, the direct path becomes computationally more "expensive". Consequently, the OSPF algorithm recalculates the topology and reroutes traffic through R2, as that alternative path now provides a lower total accumulated metric.

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

____ The Router ID acts as the unique identifier for a router within the OSPF domain. Manually controlling its assignment prevents duplicate IDs, which would otherwise disrupt neighbor establishments and corrupt the OSPF topological database. _____

2. Why is the DR/BDR election process not a concern in this lab?

____ DR and BDR elections only occur on multi-access broadcast networks. Since the routers in this specific lab topology are connected primarily via point-to-point serial links, the election process is inherently bypassed. _____

3. Why would you want to set an OSPF interface to passive?

____ Setting an interface to passive disables the transmission of OSPF Hello messages. This is highly recommended for interfaces facing end-user devices (LANs) to save bandwidth, reduce device CPU overhead, and enhance security by not broadcasting routing protocol details to untrusted networks. _____

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

____ OSPF's use of "Areas" enables its hierarchical design. By segmenting a large network into multiple smaller areas that all connect to a central backbone, OSPF isolates routing updates, minimizes routing table sizes, and greatly improves overall network scalability. _____

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

____ network 10.0.0.0 0.255.255.255 area 0 _____

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

____ The router will select the EIGRP route. When a router learns about the same destination from multiple distinct routing protocols, it relies on Administrative Distance as the tiebreaker. Since EIGRP has the lowest AD compared to OSPF and RIP, it is considered the most trustworthy source. _____